

Research Project

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Research Project

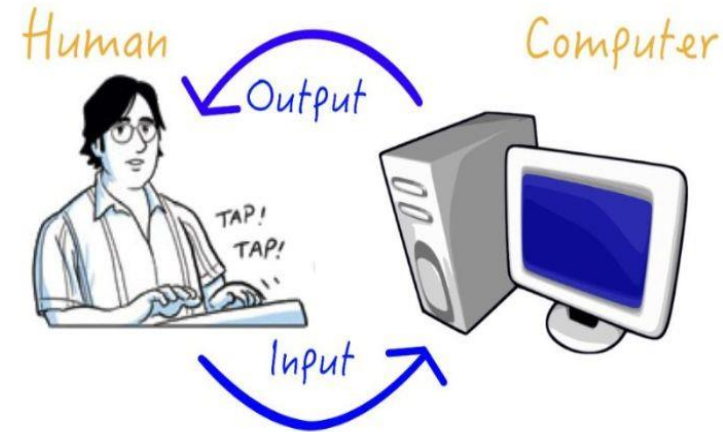
- 75% of the grade.
- Work by yourself or as a pair.
- If working as a pair, must clearly describe what each person did.
- How will the project be marked:
 - Project Proposal (10%)
 - Need confirmation.
 - Can meet and discuss.
 - Submit by end of semester: 03/07/2026.
 - Report (90%)
 - Submission time: 06/09/2026 (extensions only if a valid reason).
- Additional Details In PDF.
- GPU Resources: Use Collab. If using lab's resources, you must check with your advisor.

My Research

Scene **control**: Semantics and Physics



How to express intent? Conversation, Text, Voice, Hand Gestures



How can we **control/edit/reconstruct**:

- **Physical properties** (light, reflections, materials, etc)?
- **Semantic properties** (interactions, composition, object placement, etc)?

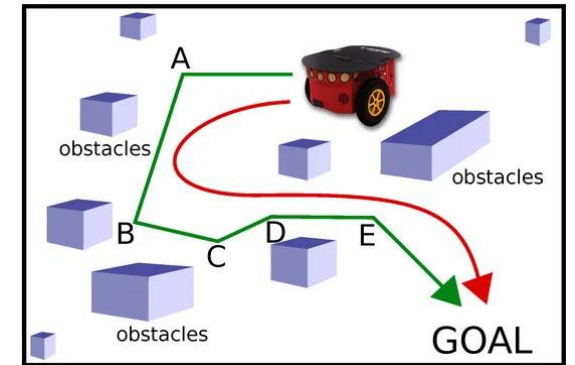
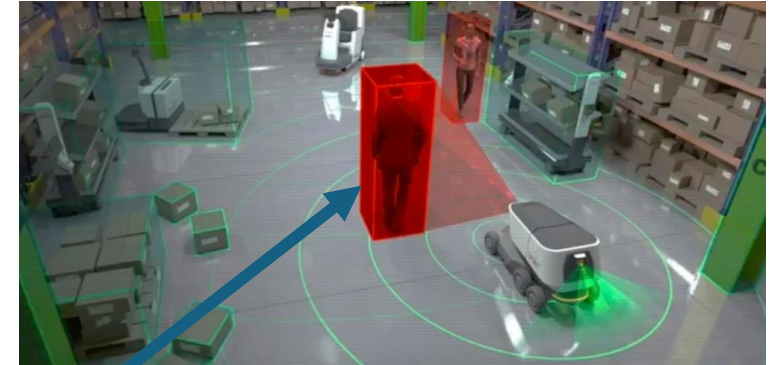
What representations do we need? **Fast, compact and accurate**

Generate Photorealistic Simulations



- How can we **disentangle** and **manipulate** objects such as people, cars, buildings, etc?
- **Infinite** diverse and photorealistic simulations.
- Generate “**hard**” **examples** at the tail of the distribution.
- **Long** Multiview Video Generation.

Understanding the world



- How can we incorporate **generative models** with **discriminative models** for perception?
- Learn these models from **few examples**:
 - Use “general” inductive biases.
 - Adapt by making analogies.
 - Distill large 2D foundation models

Slide Credit

- MIT 6.S980: Machine Learning for Inverse Graphics, Prof. Vincent Sitzmann
- MIT 6.8300/6.8301: Advances in Computer Vision, Profs Vincent Sitzmann, Bill Freeman, Phillip Isola, Antonia Torralba

Why to do research?

- Follow your interest
- Be creative
- Learn the scientific method: how to find truth through hypothesizing, predicting, experimentation.
- Discover something no-one knew before
- Safe space to be bold: tackle big problem.
- Fundamentally entrepreneurial. High risk – high reward.

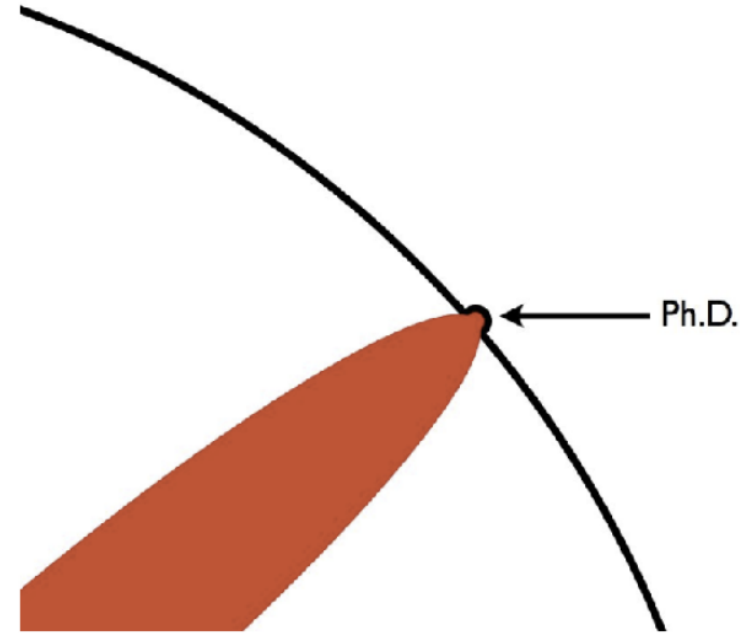
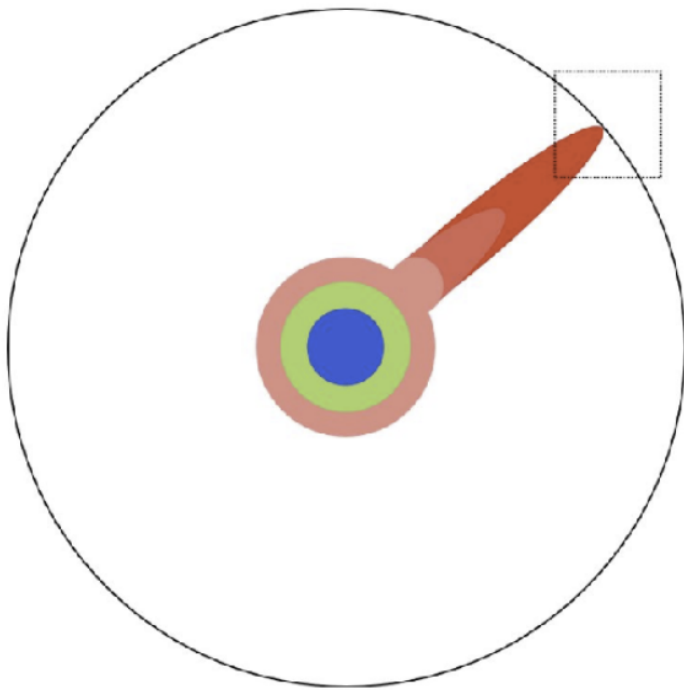
How to recognize good research?

- Novelty: Aim to work on something that is surprising.
 - People should gain a lot by reading it.
 - It should be new.
 - It should be understandable.
 - It should present idea a bit outside the current paradigm.

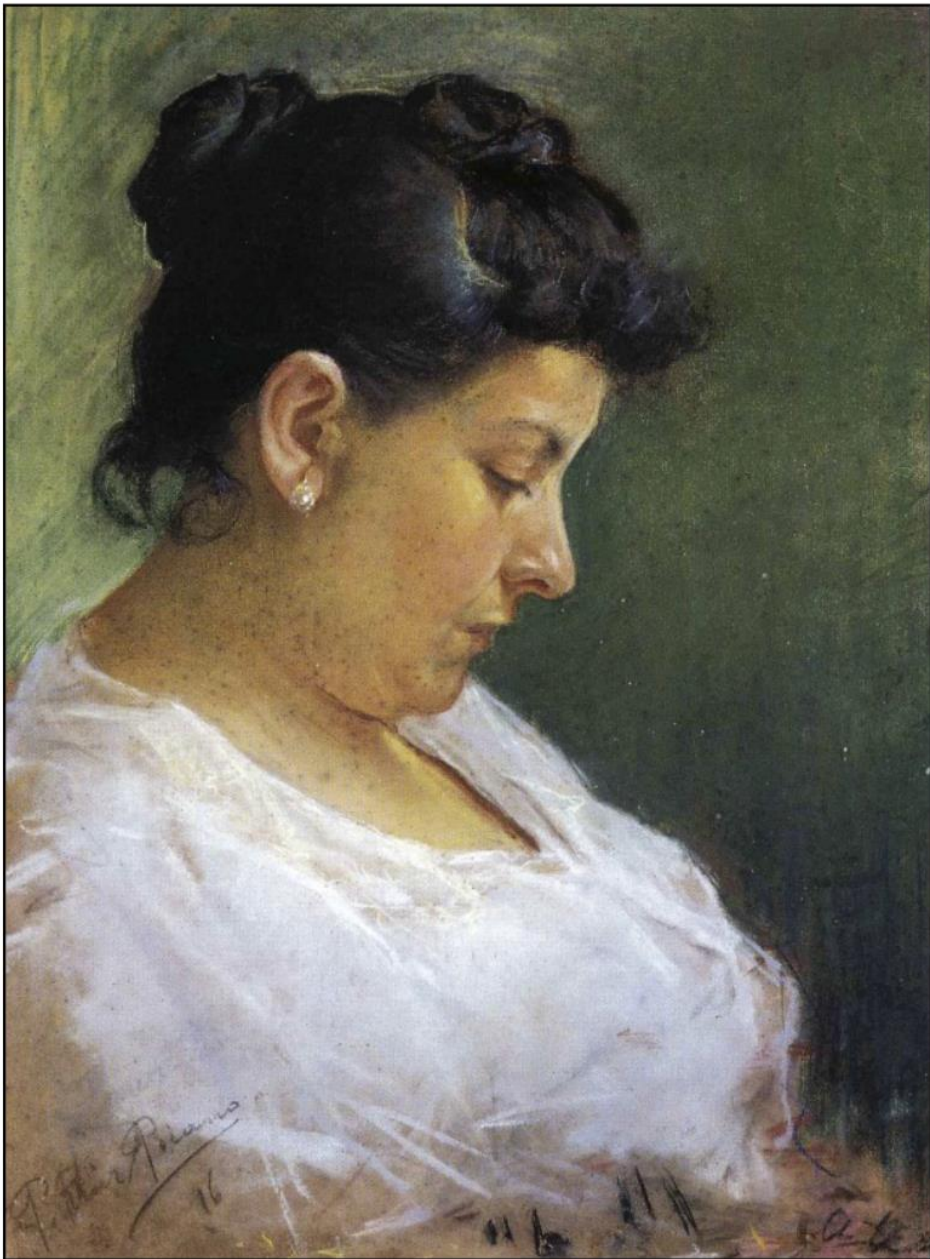
Novelty

What matters is novelty w.r.t. humanity.

Very hard to achieve without knowing what has already been done.



<http://matt.might.net/articles/phd-school-in-pictures/>



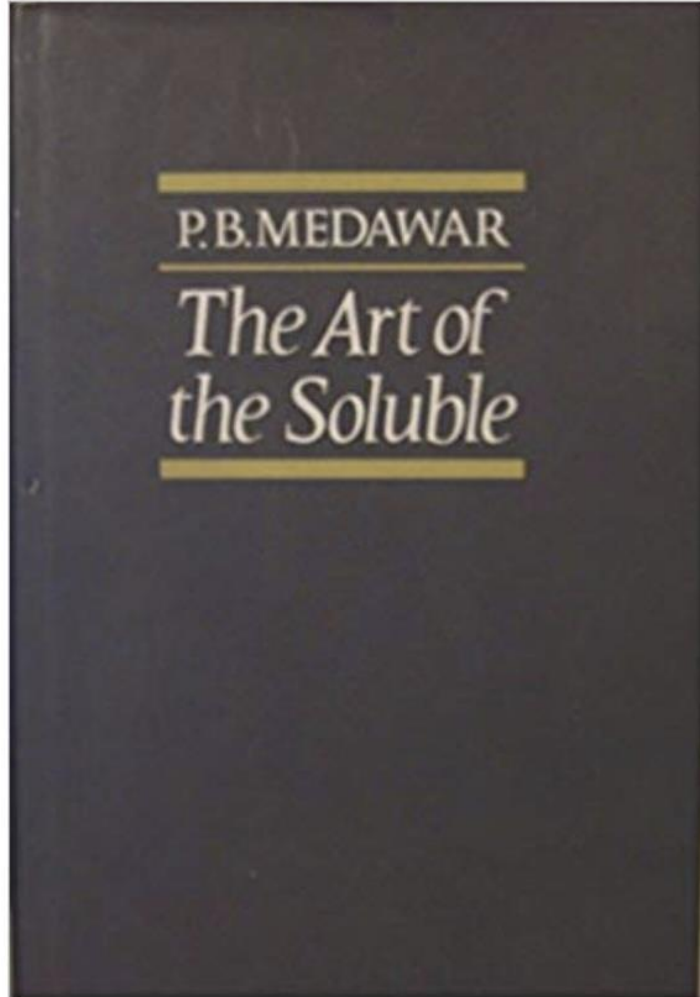
[Picasso]

How to do good research?

Picking a topic



Science is the “Art of the Soluble”



“Good scientists study the most important problems they think they can solve. It is, after all, their professional business to solve problems not to grapple with them.” —Peter Medawar”
— Jitendra Malik

How can you contribute?

- Mathematical theory.
- Empirical theory.
- Tools: Café, Pythorch (choose wisely given AI).
- Propose a new problem: needs to be motivated by a real need.
- Solving old problems better: e.g. 3D reconstruction via generation.
- Communication: Presenting ideas in a nice way, or showing a different way of looking at a problem.

The (abridged) Heilmeier Catechism

- Problem definition: What are you trying to do?
- How is it done today? What are the limits of current practice?
- What is your approach and why is it better?
- Who cares? If it is successful, what difference will it make?
- What are the “mid” and “final” exams to check for success?
- Great recipe for your project proposal!

Know something no one knows

- Necessary but not sufficient to master the core knowledge of the field.
- Acquire a new skillset
 - Take a difficult or unusual classes
 - Read old papers
 - Take on a complementary hobby
 - Talk to people in other fields

“To be a good scientist, you need a secret superpower. You need to do something better than everybody else. The great thing about science is that we don’t all have the same superpower. Maybe my superpower has been that, because I don’t see very well, I might have more insight into the vision problem.”

Alexei Efros.

Question obvious weirdness

- Often, when you are new to the field, you are confronted with something everyone accepts, but that strikes you as weird.
- These are often good research topics.

Think about the consequences

- Why are you working on the problem you are working on?
- What would happen if you are successful? Is that what you want to happen? Consider impact.

Get comfortable being confused

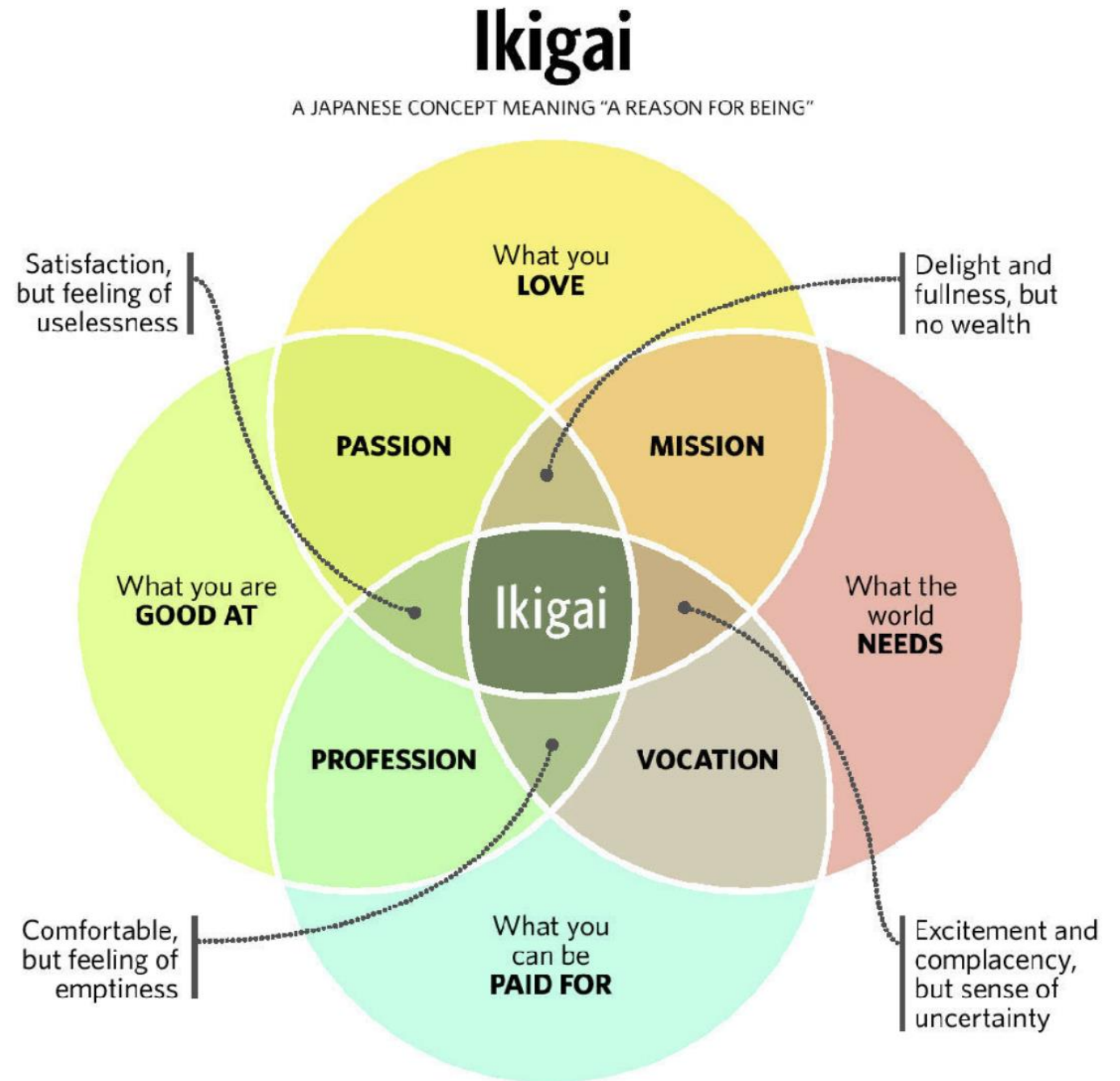
“Great scientists tolerate ambiguity very well. They believe the theory enough to go ahead; they doubt it enough to notice the errors and faults so they can step forward and create the new replacement theory. If you believe too much you'll never notice the flaws; if you doubt too much you won't get started.”

Richard W. Hamming.

Picking a topic

Don't worry too much if a topic is unpopular

If it is important, and you do good work on it, you can *make it popular*



Omit needless bits

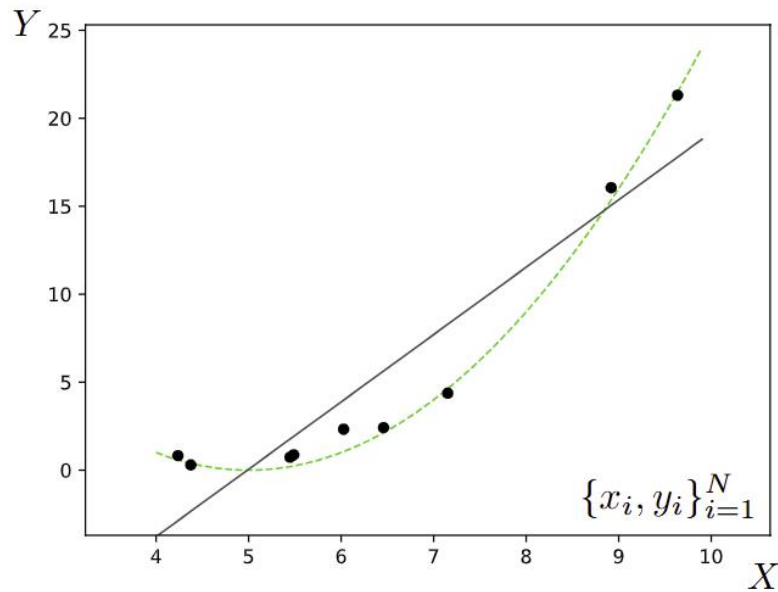
"Perfection is finally attained not when there is no longer anything to add, but when there is no longer anything to take away"

— Antoine de Saint Exupéry



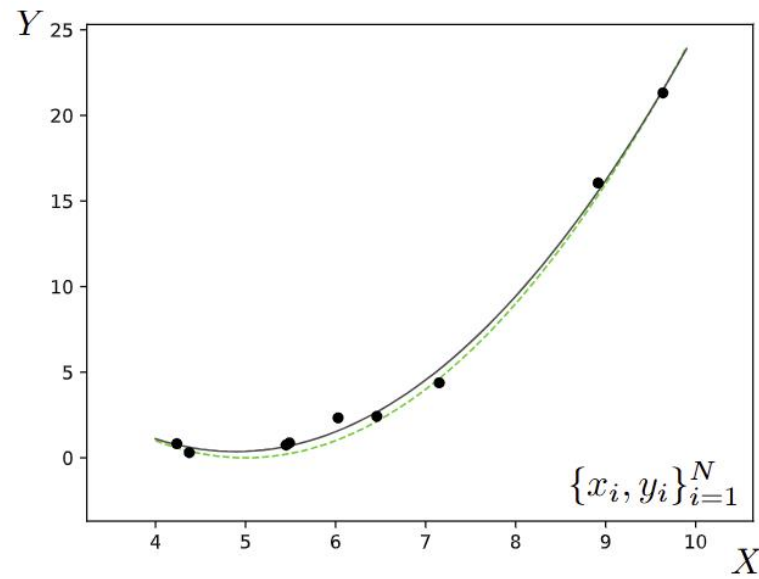
Underfitting

$$K = 1$$



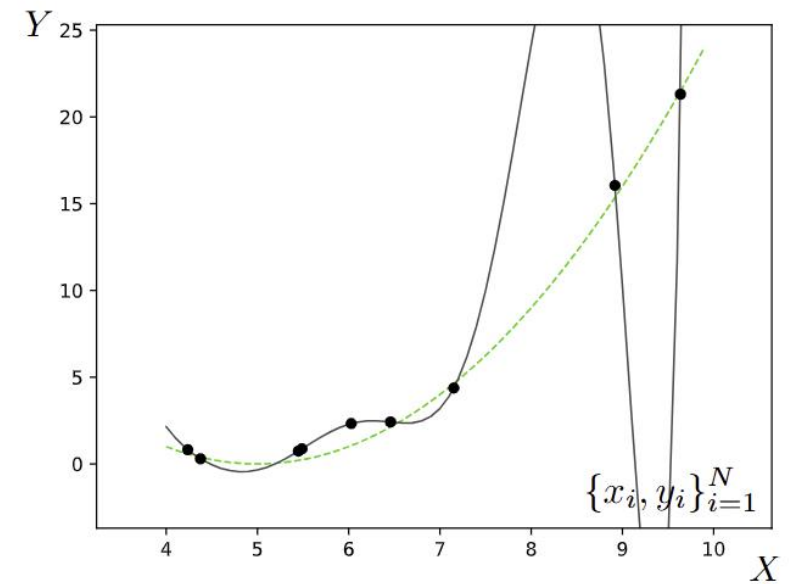
Appropriate model

$$K = 2$$

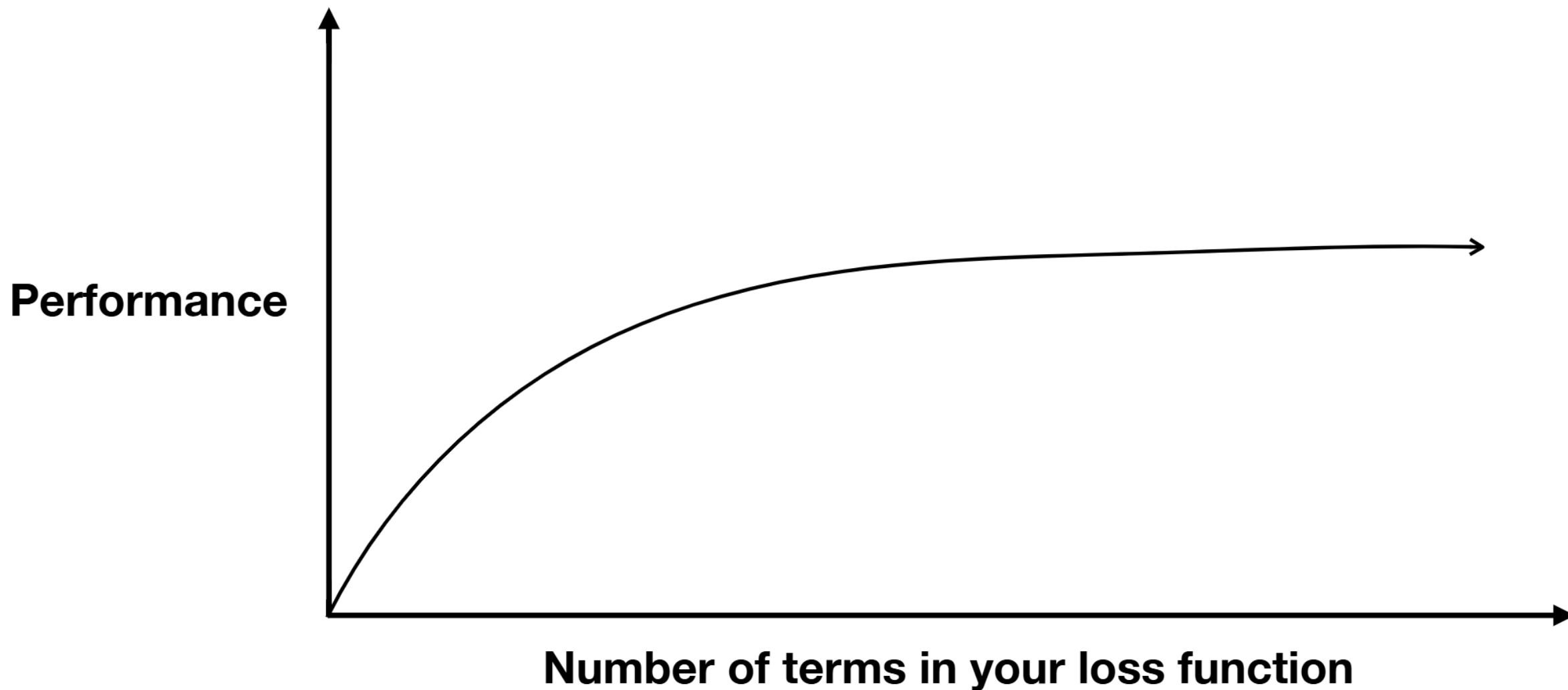


Overfitting

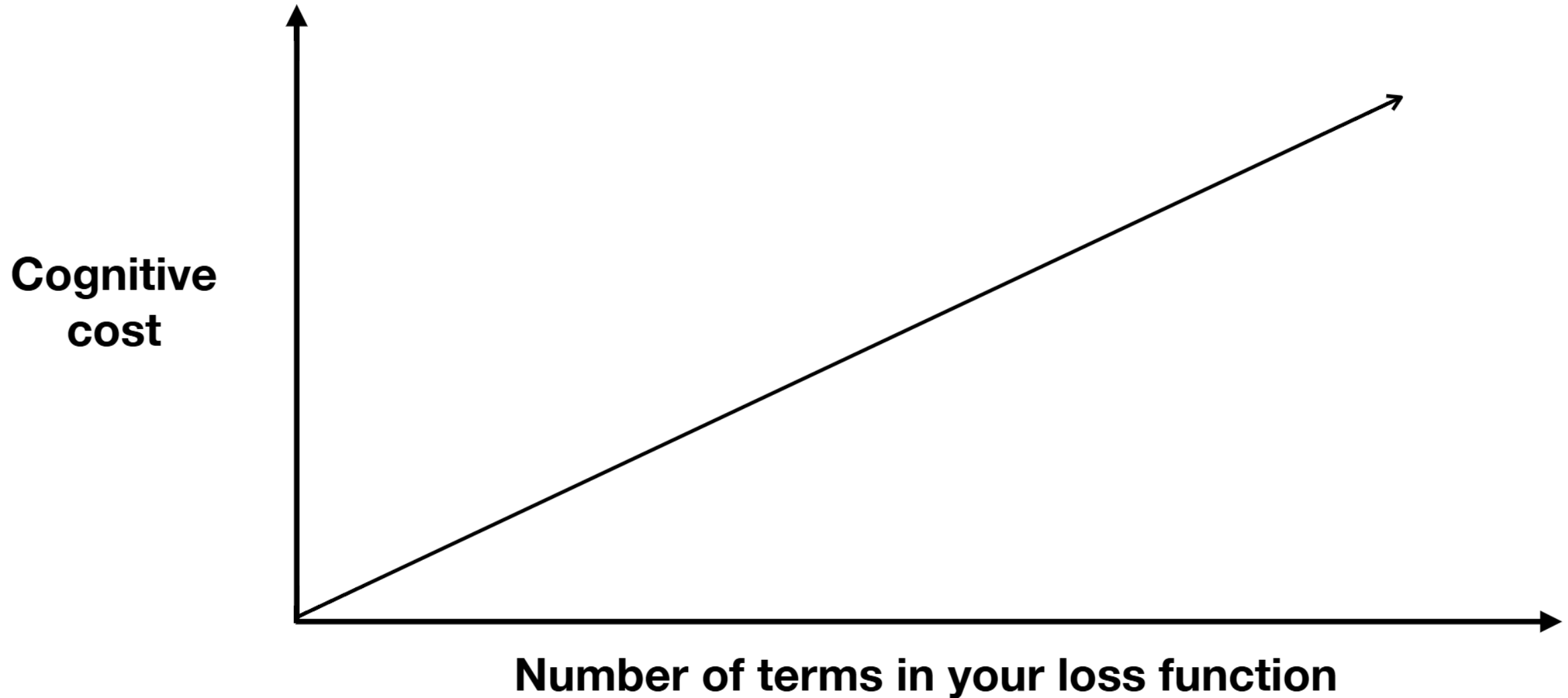
$$K = 10$$



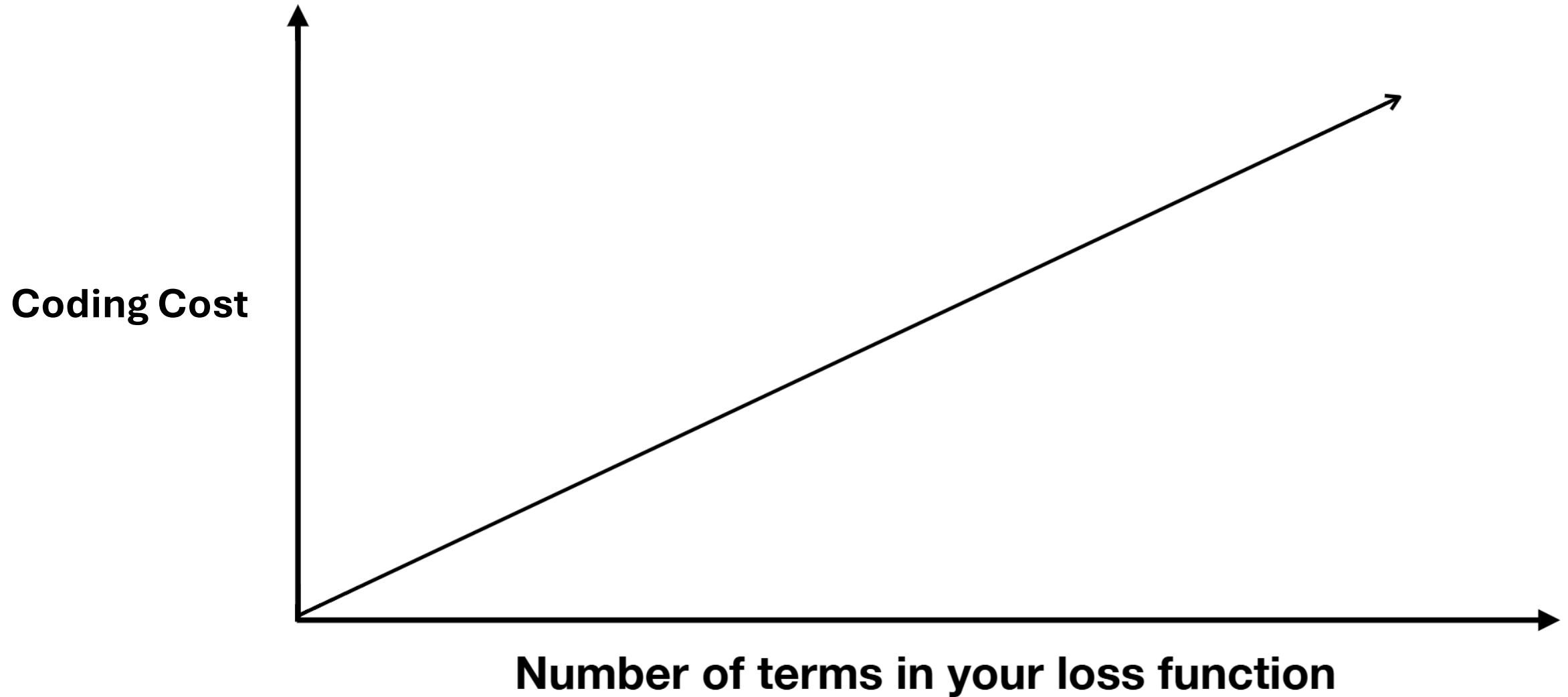
Information creep



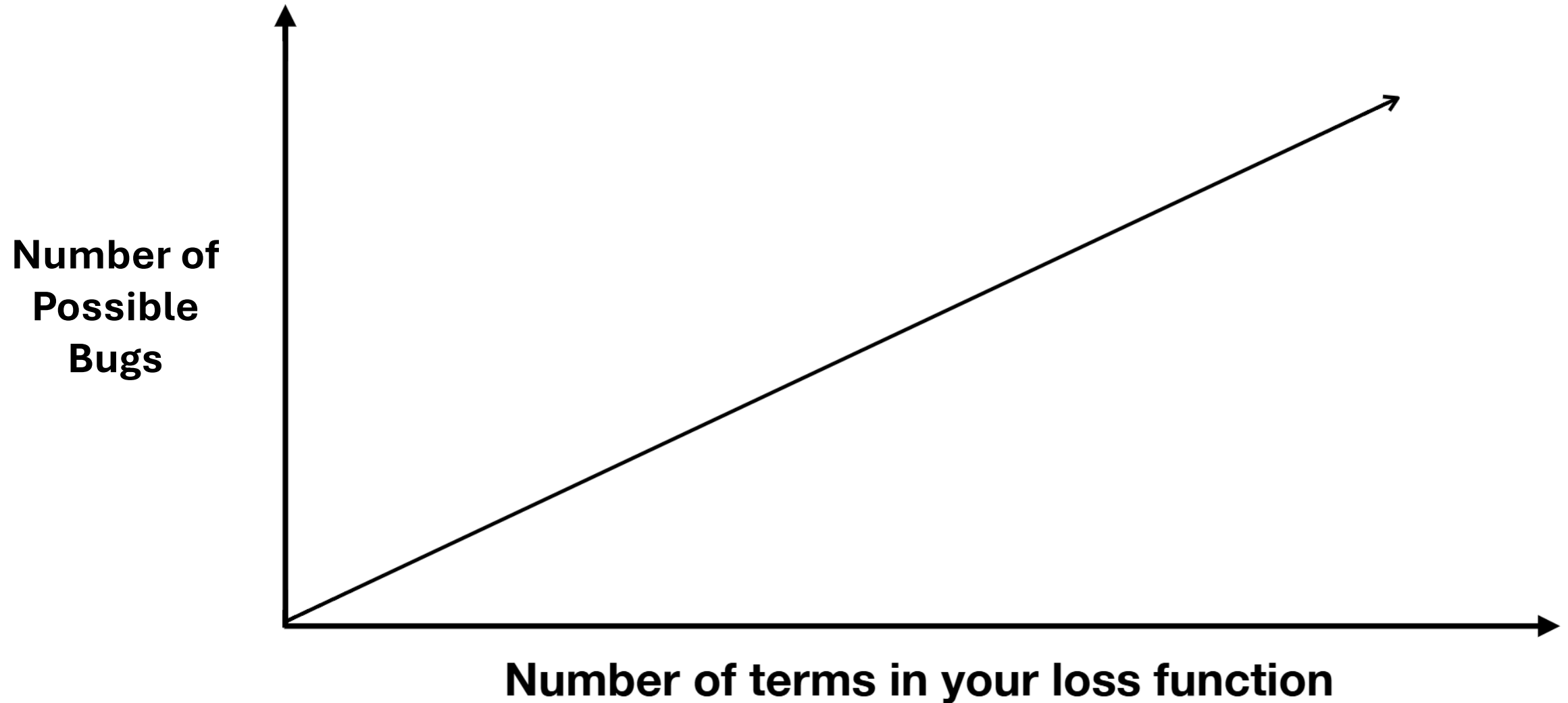
Information creep



Information creep



Information creep



Do the most, with the least

Products

Discoveries
Explanations
Results
Tools

...

=

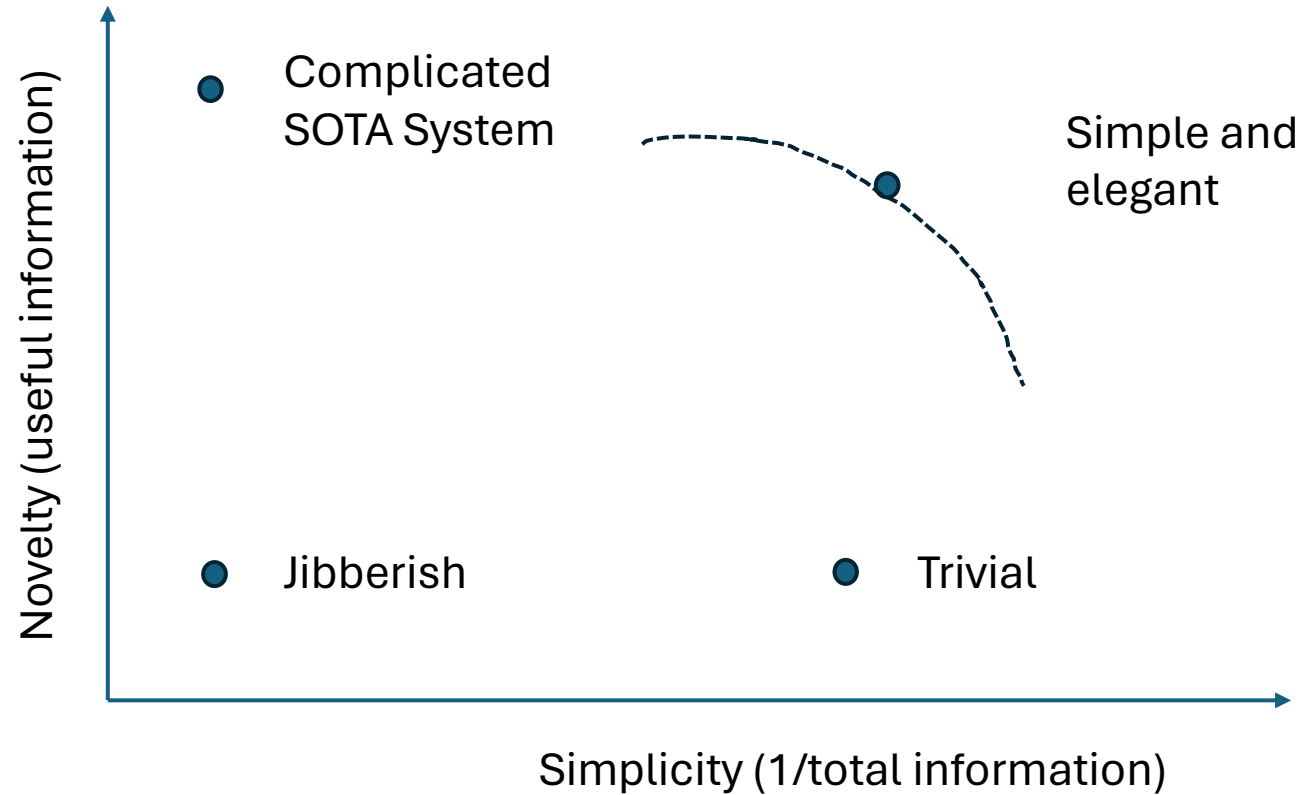
Metric for research

Costs

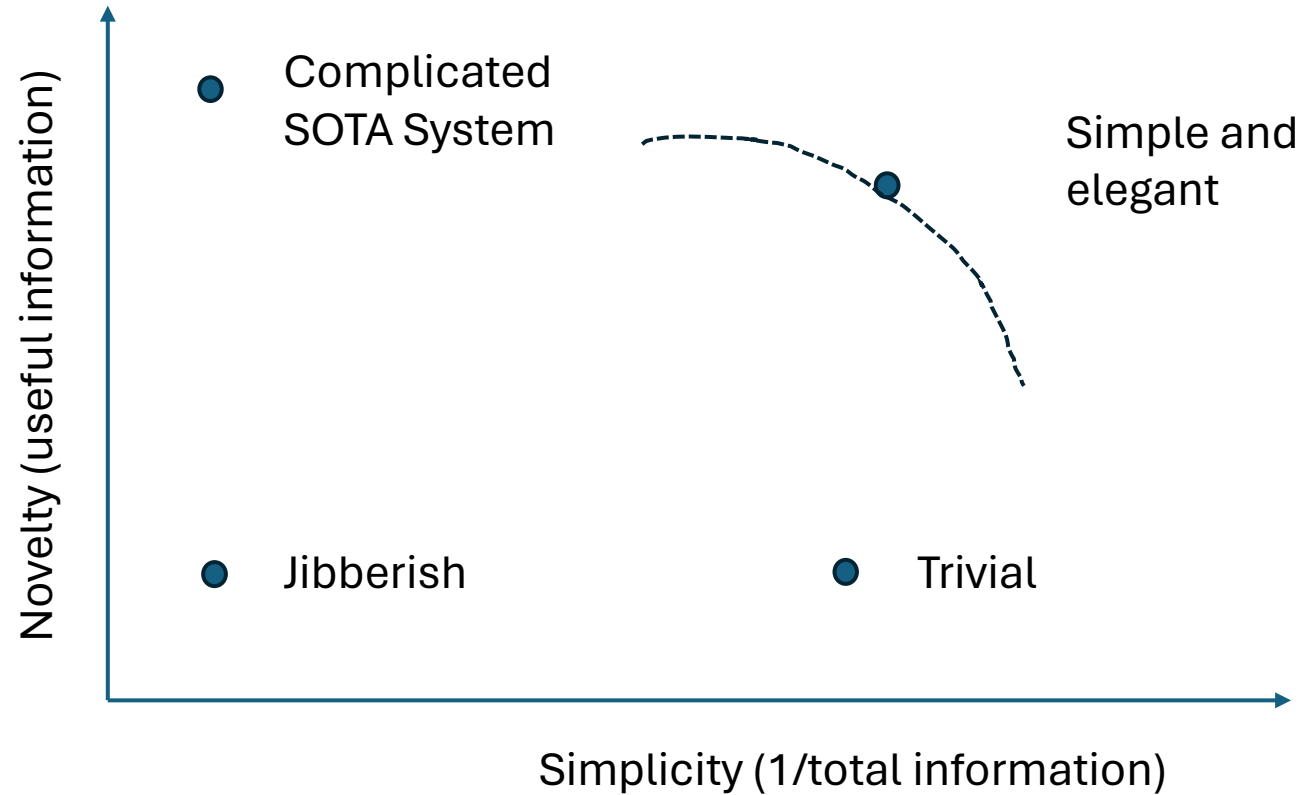
Words
Equations
Concepts
Lines of code
GPUs
People, Time, Money

...

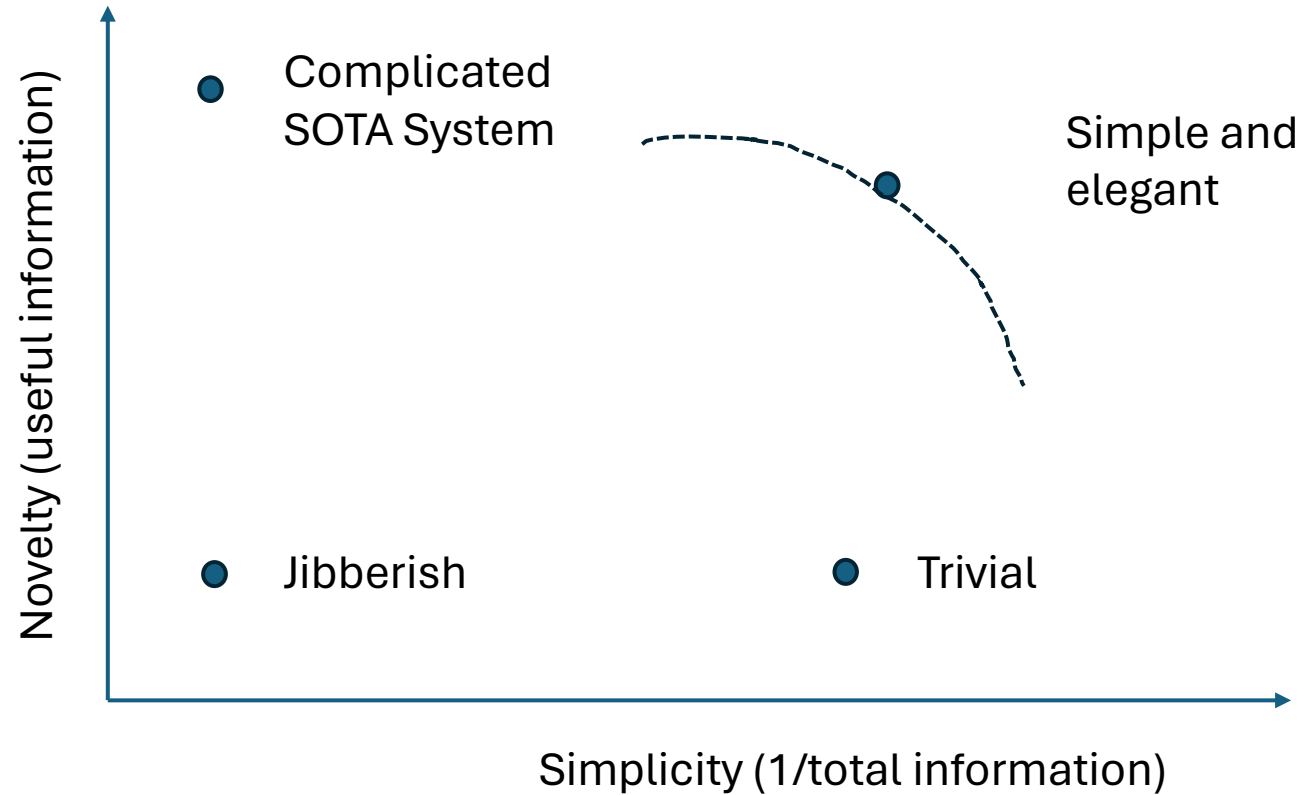
Pareto front of simplicity vs novelty



What reviewing system rewards?



What gets test-of-time award?



Scope your contribution

- Pick the broadest scope under which your work is novel. Too broad and your work is indistinguishable from prior theory. Too narrow and your work is not impactful.

The regularizing force of human fallability

- The vast majority of information at any given academic conference is forgotten – that's a good thing.
- We tend to forget all but simplest and starkest discoveries.
- Complex and subtle discoveries are usually either overfit or unimportant

Other Resources

- How to do research by Bill Freeman:
https://people.csail.mit.edu/billf/publications/How_To_Do_Research.pdf
- Watching debates between scientists.
- Watch tutorial and workshops and CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, etc from recent years.
- Follow the thread of a given topic: <https://www.undermind.ai/> - Great Tool!
- Use Scholar, etc.